Lab Practical Assignment

Case Study

A financial institution maintains customer accounts and charges a monthly maintenance fee of Ksh. 200 on every active customer account at the end of each month.

Currently, the accounts department updates customer balances manually. This process is time-consuming and may result in errors or failure to deduct the monthly maintenance fees.

The institution wants to automate this process using SQL triggers. Whenever a monthly account update is performed, the system should automatically deduct the maintenance fee from each eligible customer account and record the transaction.

2. Lab Objectives

By the end of this practical, students should be able to:

1. Create database tables for customer account management.
2. Write SQL triggers to automate database operations.
3. Use triggers to maintain accurate customer account balances.
4. Automatically record account transactions.
5. Test and evaluate the functionality of SQL triggers.

3. Database Setup

Create a database called `CustomerManagementDB`.

Task 1: Create the database tables (3 Marks)

Create the following tables:

A. Customers table

Field	Data type	Description
customer_id	INT	Primary key
customer_name	VARCHAR(100)	Customer's full name
account_number	VARCHAR(20)	Unique account number
balance	DECIMAL(10,2)	Current account balance
status	VARCHAR(20)	Active or Inactive

B. Transactions table

Field	Data type	Description
transaction_id	INT	Primary key, auto increment
customer_id	INT	Customer ID
transaction_type	VARCHAR(50)	Type of transaction
amount	DECIMAL(10,2)	Transaction amount
transaction_date	DATETIME	Date and time of transaction

Task 2: Insert sample customer records (2 Marks)

Insert at least five customers into the `Customers` table using the following sample data.

Customer ID	Customer Name	Account Number	Balance (Ksh.)	Status
1	John Kamau	ACC001	15,000	Active
2	Mary Wanjiku	ACC002	8,000	Active
3	Peter Otieno	ACC003	5,000	Active
4	Grace Akinyi	ACC004	12,000	Inactive
5	James Mwangi	ACC005	100	Active

4. Trigger Development

Task 3: Create an SQL trigger (4 Marks)

Develop a database trigger that automatically updates customer accounts at the end of every month.

Your trigger should:

1. Deduct a monthly maintenance fee of Ksh. 200 from every active customer account.
2. Ensure that inactive accounts are not charged.
3. Ensure that accounts with balances below Ksh. 200 are not charged.
4. Automatically insert a record into the `Transactions` table whenever a maintenance fee is deducted.

Important: The trigger should execute automatically when a monthly account update is initiated.

Hint: Consider using an `AFTER UPDATE` trigger on the `Customers` table or an appropriate trigger on a separate monthly processing table.

5. Testing and Verification

Task 4: Test the trigger (3 Marks)

Perform the following activities:

1. Execute the monthly account update process.
2. Display all customer account balances after the update.
3. Display all transactions recorded in the `Transactions` table.
4. Verify that inactive accounts and accounts with insufficient balances have not been charged.

Use appropriate SQL `SELECT` statements to demonstrate your results.

6. Evaluation and Critical Thinking

Task 5: Short-answer questions (3 Marks)

Answer the following questions:

a) Explain why a trigger is useful in automating monthly customer account updates. (1 mark)

b) Differentiate between a `BEFORE UPDATE` trigger and an `AFTER UPDATE` trigger. (1 mark)

c) Explain one limitation of using a trigger to perform monthly account updates. (1 mark)